

CrowdVision — Role Assignment (3 Technical + 1 Ops)

Companion to Build Plan v9. Print this page or keep it open on a phone. Every hour, every person knows exactly what they're doing without consulting the 600-line plan.

The Four Roles

ALPHA — The Brain (strongest developer)

Owns: the entire zone-brain on the Surface Laptop — everything the NPU touches and everything that makes safety decisions.

Permanent ownership	What this means concretely
Vision pipeline	<code>capture.py</code> (5-feed RTSP + file ingest, watchdog), <code>scheduler.py</code> (round-robin freshest-frame), <code>detect_qnn.py</code> (shared QNN session), <code>homography.py</code> , <code>density.py</code>
Tracker + gate lines	<code>tracker.py</code> (centroid tracker), <code>gatelines.py</code> (real lines on C4 feed, virtual lines on zone views)
Risk engine	<code>risk.py</code> (EWMA, TTT, flow conservation, hysteresis, stale-feed policy), <code>playbooks.py</code>
NPU verification	<code>verify_npu.py</code> — runs it first thing Saturday, owns the proof artifact
Benchmarks (vision)	<code>detect_bench.py</code> , <code>mesh_bench.py</code> (5-feed sustained soak), <code>power_delta.ps1</code>

Alpha does NOT touch: the gate node, the phone app, the dashboard frontend, or any documentation file. If Alpha is debugging the QNN session at 2 a.m., nothing else should interrupt that.

Pre-event (July 8–10): kicks off AI Hub exports (YOLO x2, Genie bundle in WSL2), assembles the 60-image calibration set, writes the full vision pipeline and scheduler against file feeds on the personal laptop, runs the mesh dress rehearsal (July 10, 60 min, 4 RTSP + 1 file through the pipeline).

BETA – The Hands (comfortable with both Android/Kotlin and Arduino/C++)

Owns: every physical device that isn't the Surface – the UNO Q gate node and the OnePlus officer app. Beta is the person whose work judges can *touch and watch*.

Permanent ownership	What this means concretely
Gate node (UNO Q)	<code>gate-node/</code> – the full App Lab app: <code>sketch.ino</code> (Bridge RPC: matrix patterns, RGB, Knob/Buzzer/Thermo behind feature flags), <code>main.py</code> (MQTT client, fail-safe state machine, LWT logic)
Officer app (OnePlus)	<code>field-app/</code> – Kotlin + Compose: MQTT dispatch receive/ack, GPS beacon (AOSP LocationManager), incident reporting (FunctionGemma 270M integration + dropdown form fallback), Officer-2 instance on Phone-H
Bridge RPC	The single highest-risk integration – Beta owns de-risking it pre-event and being the first to test it on real hardware Saturday 13:00
E2B NPU probe	Saturday 13:30 timeboxed 30-min probe on the OnePlus: load the model, capture result, done – Beta then returns to the officer app
Benchmarks (devices)	FunctionGemma TTFT/tok-s, Bridge RPC round-trip, gate actuation timing

Beta does NOT touch: the vision pipeline, the risk engine, the dashboard, or docs. Beta's hands are literally on the UNO Q and the phone during the demo.

Pre-event (July 8–10): downloads FunctionGemma + E2B model + NPU `.so` set; builds the Kotlin officer app skeleton (MQTT, dispatch, beacon, report, form, E2B probe flag behind a toggle), `assembleRelease` → APK; writes the full gate-

node App Lab app; pins Bridge API names from the User Manual + built-in examples (guide warns they vary by version); runs the mocked-Bridge test; installs IP Webcam on all camera phones (C1–C4), presets 640×480 @ 12 fps.

GAMMA — The Glue (full-stack: Python backend + JS frontend + cloud integration)

Owens: everything that makes the system *visible and connected* — the MQTT broker, the dashboard, the cloud venue tier, the camera mesh config, and the sim harness.

Permanent ownership	What this means concretely
MQTT broker	<code>mosquitto</code> setup on the Surface (or <code>amqtt</code> fallback); topic structure; LWT configuration for every device
Dashboard	<code>zone-brain/server/</code> — FastAPI + WebSocket fan-out, Leaflet local-CRS floorplan, feed-health chips, zone/gate/officer/incident rendering, event log with decision provenance , per-gate operator override buttons, backend+latency badges on every message
Cloud venue tier	<code>venue-tier/</code> — AI Inference Suite REST client, trilingual advisory prompts, sim-zone publisher (2 SIM zones), <code>template_fallback.py</code> , Sarvam adapter slot
Camera mesh config	<code>config/cameras.yaml</code> (RTSP URLs, resolution, fps), <code>config/devices.yaml</code> (IPs), <code>config/zones.yaml</code> (polygons, thresholds) — Gamma writes the configs, Alpha's pipeline reads them
Calibration tool	<code>tools/calibrate.py --camera c1..c4 --verify</code> — Gamma runs it Saturday with Delta positioning phones
Sim harness	<code>sim/</code> — <code>sim_feeds.py</code> (5 looping file feeds), <code>sim_gate.py</code> , <code>sim_officer.py</code> , <code>sim/tests/</code>
Benchmarks (network)	<code>net_bench.py</code> (hotspot throughput, RTSP drop rate), <code>e2e_bench.py</code> (frame → gate round-trip), Cloud AI 100 RTT
Attendee QR view (stretch)	<code>server/attendee.py</code> — if time exists post-G4, Gamma builds it

Gamma does NOT touch: the vision pipeline internals, the QNN session, the phone app, or the gate node sketch. Gamma's integration testing is done via MQTT messages — if the right JSON arrives on the right topic, Gamma's side works regardless of what produced it.

Pre-event (July 8–10): builds the full dashboard + broker + sim harness; gets `--sim-all` working end-to-end (the judges' install path — this must be done by July 10); writes all config files; builds the venue-tier client; tests hotspot throughput with the real phone mesh (July 10 dress rehearsal with Alpha).

DELTA — Story & Ops (non-technical; the conscience)

Owens: everything that isn't code — which is worth **35 points** (Innovation framing 25 + Presentation 15, minus the parts that are inherent in the code). Delta is the reason the team doesn't lose points to exhaustion, missed deadlines, or a fumbled demo.

Permanent ownership	What this means concretely
The clock	Announces every gate (G0–G6) on time, calls the pre-agreed fail branch if the gate is red, enforces the two-strike bug rule, calls feature freeze at G4 03:00. Delta's authority on gates is absolute — no "five more minutes."
Demo narration	Primary Narrator during the 5-minute demo. Knows the script cold. Rehearses ×2 Sunday morning. Owns the 3:30 cue card (drop-dead jump to Numbers). Delivers the hook, the kill-shot punchline, and the close.
Documentation	README (from the template, filled), ARCHITECTURE.md, MESSAGES.md, DEMO.md, DEVICES.md, THIRD_PARTY_LICENSES.md. Writes them in real-time as features land — never as a 6 a.m. panic.
BENCHMARKS.md	Runs the bench scripts (they're one-command), collects the JSON output, embeds the tables. Delta doesn't write the scripts — Alpha/Beta/Gamma do — but Delta runs them and owns the doc.
Compliance	§n + §o checklists: all five names+emails (including the 4th member), MIT license, no sponsor marks, no social media posts (§17.c), scope-confirmation email, loaner agreement,

Permanent ownership	What this means concretely
	Form submission by 12:15. Delta screenshots every compliance action.
Physical ops	Modulino sprint at 12:00 distribution (be first in line); camera phone positioning + charging audits (17:00, 23:00, 02:30, 08:30); "GATE 3" card folded from venue paper; demo station reset Sunday 12:15.
Fresh-clone test	Sunday 06:00: follows the README cold on a personal laptop — the test is whether a <i>non-author</i> can reproduce the setup. Fixes go to docs, not code. This is Delta's most important morning action.
Backup video	Sunday 07:00: records the 2-minute screen capture of the full live loop; stores on the Surface desktop + a personal phone.
Sarvam listener	Attends the 11:30–12:00 Sarvam session; captures any API/key/model offer; reports to Gamma at lunch for the G2 adopt/skip decision.
Q&A prep	Holds the three pre-rehearsed answers (cloud-only? occlusion? operator acceptance?) + the energy answer + the camera-realism answer. If a judge asks during the demo and the Narrator stumbles, Delta can interject.

Delta does NOT: write code, debug code, touch the terminal, or make technical decisions. If Delta is coding at any point, the plan has failed. Delta's value is being the fresh eyes, the clock, and the voice.

Schedule Remapped to 3+1

Time	ALPHA (Brain)	BETA (Hands)	GAMMA (Glue)	DE
Sat 10:00	Check-in	Check-in	Check-in	Cr co (g re as

Time	ALPHA (Brain)	BETA (Hands)	GAMMA (Glue)	DE
11:00–12:00	Kickoff + DevRel masterclass (all attend)	Kickoff	Kickoff	Sa de
12:00–13:00	Lunch	Lunch	Lunch	Sp tal Kr Le
13:00–14:00	Surface: wheelhouse install → <code>verify_npu.py</code> (NPU proof)	UNO Q first boot: Blink + Bridge RPC echo (the highest-risk moment) → OnePlus APK install	Phone-H hotspot up; broker started; camera phones configured (with Delta positioning); RTSP URLs into <code>cameras.yaml</code> ; all feeds visible in <code>capture.py</code>	Ce (le pli 10
13:30–14:00	(continues NPU verification)	E2B NPU probe (30-min timebox, hard stop 14:00; result recorded either way)	(continues mesh config)	Ar
14:00–17:00	Feed A + C1 through shared QNN session; per-frame ms measured; add remaining feeds to scheduler	Gate state machine: MQTT subscription → Bridge RPC → matrix/RGB patterns; Modulino feature flags	Dashboard rendering: floorplan + fake zones + feed-health chips; broker topic structure	Pe Ge ca — cli dr
17:00	—	—	—	Ar NF flic
17:00–20:00	Homography per camera → density → tracker → real gate lines	Fail-safe timer + knob override + buzzer chirp (if secured)	Zones/gates/officers/incidents live on dashboard + event log with provenance fields	Be th ch

Time	ALPHA (Brain)	BETA (Hands)	GAMMA (Glue)	DE
	(C4) + virtual lines → risk states			ch st:
19:30–20:15	Dinner (shift 1 with Beta)	Dinner (shift 1 with Alpha)	Dinner (shift 2 with Delta)	Dii
20:00–21:00	Integration: send density → sim gate → dashboard, full loop in sim	Integration: officer app MQTT wiring; Officer-2 on Phone-H	Integration: cloud advisory call + venue view (2 SIM zones) + uplink-cut handling	—
21:00	—	—	—	Ar sir Cæ
21:00–24:00	TTT predictor + flow conservation + hysteresis + stale-feed policy + playbooks P1/P2/P3	Real gate end-to-end — physical LEDs flip from live density; nearest-dispatch + ack chain	Cloud advisory wiring; template fallback; (Sarvam if adopted — 90 min timebox ends ~22:30)	MI (di cli ov
24:00	Midnight rehearsal (all): one full live run, rough stopwatch	Midnight rehearsal	Midnight rehearsal	MI th nu
00:30	—	—	—	Ar p9 Of co
00:30–03:00	Should-have (fall-confirm) OR hardening (reconnects, watchdog tuning)	Should-have (Thermo/Buzzer wiring) OR hardening (Bridge robustness)	Should-have (replay heatmap) OR hardening (QoS audit, badge polish)	RL ve m e2 en BE au

Time	ALPHA (Brain)	BETA (Hands)	GAMMA (Glue)	DE
03:00	—	—	—	Ar FF dc
03:00–06:00	Sleep pod 1 (03:00–04:30) → watch + soak test monitoring (04:30–06:00)	Sleep pod 1 (03:00–04:30) → sleep pod 2 (04:30–06:00)	Watch + soak test (03:00–04:30) → sleep pod 2 (04:30–06:00)	Sl ne na st:
06:00–07:00	Final vision-side bug fixes (demo-critical only, two-strike rule)	Final device-side bug fixes	Final dashboard/cloud fixes	Fr RE lap Re ov
07:00–08:00	Final benchmarks: 300-frame NPU vs CPU, 5-feed aggregate, battery deltas	FunctionGemma TTFT/tok-s ×20, Bridge RPC ×100	e2e ×50, hotspot throughput, Cloud RTT ×30	Re (s loc pe au ph
08:00–09:00	Benchmark results → JSON committed	Benchmark results → JSON committed	Benchmark results → JSON committed; BENCHMARKS.md auto-embedded	RE ev 4 i TH dc
09:00	—	—	—	Ar re: sc pc co
09:00–10:00	Demo rehearsal #1 (timed — Alpha drives)	Demo rehearsal #1 (Beta at the gate, holds OnePlus)	Demo rehearsal #1 (Gamma ensures dashboard is on-screen)	De na 3:.
10:00–11:00	Demo rehearsal #2 + feed-mix	Demo rehearsal #2	Demo rehearsal #2	De "G

Time	ALPHA (Brain)	BETA (Hands)	GAMMA (Glue)	DE
	lock: C2/C3 live only if zero dropouts across both rehearsals			pa
11:00–11:45	Repo freeze: <code>git tag v1.0</code> , push	Verify APK in Releases downloads	Verify <code>--sim-all</code> from the public repo on a phone in incognito	Ve MI log
11:45–12:15	—	—	—	SL Sc Ne Ar
12:15–13:00	Reset demo station; cache one cloud advisory response as slide backup	Reposition UNO Q + phone; top up charges	One silent end-to-end dry run	Pl cu
13:00–16:00	DEMO: Driver — operates the Surface, starts the surge clip, scrolls the log, clicks overrides, shows the numbers slide	DEMO: Device hands — stands at Gate 3, turns the Knob (or points at the override), holds the OnePlus for the officer beat	DEMO: Dashboard monitor — ensures the screen shows the right view at the right beat; switches to venue view for beat 4; toggles Phone-H data for the uplink-cut	DE th ca an
16:00	Device return (Surface)	Device return (UNO Q, OnePlus, Modulinos)	—	Lo de

Decision Authority

Decision	Who decides	Rule
Gate pass/fail (G0–G6)	Delta	Delta's call is final. No "five more minutes."
Technical triage (which bug to fix, which workaround to take)	Alpha (vision/engine) · Beta (devices) · Gamma (dashboard/cloud) — each in their domain	Two 20-min attempts, then the pre-planned workaround. If they disagree across domains, Alpha breaks the tie (the core is on the PC).
Feature freeze exception after G4	Nobody. G4 is a lock, not a suggestion.	Bugs and docs only.
Feed-mix lock (which cameras stay live)	Alpha + Delta jointly after rehearsal #2	Minimum live = Feed A + C1 + C4. C2/C3 live only if zero dropouts in both rehearsals.
Demo script changes	Delta	The script is rehearsed; ad-libs at 4 a.m. don't make it in.
Scope-confirmation wording to organizers	Delta (non-technical framing is Delta's job)	Alpha reviews for technical accuracy before sending.
Sarvam adopt/skip at G2	Gamma (owns the integration) with Delta's clock authority on the timebox	Adopt only if Gamma estimates ≤90 min AND the core (G2) is green.

One-Line Mantras (tape to the table)

- **Alpha:** "The NPU session is sacred. Nothing interrupts the vision pipeline."
- **Beta:** "Bridge echo first. If the gate doesn't flip, nothing else matters."
- **Gamma:** "If the dashboard doesn't show it, it didn't happen."

- **Delta:** "I own the clock, the voice, and the checklist. I never touch the terminal."